

HelperLearner - API Documentation

1. Overview

HelperLearner REST API provides JSON endpoints for mobile apps, external integrations, and client applications. All endpoints are RESTful and follow HTTP conventions.

Base URL: `https://helperlearner.com/api/v1/` **Content-Type:** `application/json` **Authentication:** Session-based or API Key

2. Authentication

2.1 Session Authentication

- Standard Django session auth (cookies)
- CSRF token required for POST/PUT/DELETE requests
- Session timeout: 30 days

2.2 API Key Authentication

Header: `Authorization: Api-Key <key>`

- Create API key in user dashboard → Integrations
- Keys are hashed in database (never shown again)
- Regenerate key if compromised

2.3 Rate Limiting

- **Anonymous users:** 100 requests/day
- **Authenticated users:** 1,000 requests/day
- **Premium users:** 10,000 requests/day

Response Headers:

`X-RateLimit-Limit: 1000`
`X-RateLimit-Remaining: 950`
`X-RateLimit-Reset: 1234567890`

3. Help Requests API

3.1 List Help Requests

Endpoint: `GET /requests/`

Query Parameters:

- `page`: integer (default: 1)
- `page_size`: integer (default: 20, max: 100)
- `status`: string (open, claimed, resolved)
- `difficulty`: string (easy, medium, hard)
- `min_kp`: integer

- max_kp: integer
- tags: string (comma-separated)
- sort_by: string (created, kp_bounty, difficulty)
- search: string (full-text search)

Request:

```
curl -H "Authorization: Api-Key your-key" \
  "https://helperlearner.com/api/v1/requests/?status=open&difficulty=medium&sort_by=-created_at"
```

Response:

```
{
  "count": 156,
  "next": "https://helperlearner.com/api/v1/requests/?page=2",
  "previous": null,
  "results": [
    {
      "id": 1,
      "requester": {
        "id": 5,
        "username": "john_dev",
        "trust_score": 75.5
      },
      "title": "Django ORM Query Optimization",
      "description": "How to optimize N+1 queries in Django...",
      "kp_bounty": 150,
      "difficulty": "medium",
      "status": "open",
      "tags": ["django", "orm", "performance"],
      "created_at": "2024-05-10T14:30:00Z",
      "claimed_by": null,
      "claimed_at": null,
      "ai_summary": "User seeks help with optimizing Django ORM queries...",
      "views_count": 42,
      "solutions_count": 0
    }
  ]
}
```

3.2 Create Help Request

Endpoint: POST /requests/

Request Body:

```
{
  "title": "How to implement JWT authentication in Django?",
  "description": "I need help setting up JWT for my API. Currently using basic auth...",
  "kp_bounty": 200,
  "difficulty": "medium",
}
```

```
    "tags": ["django", "jwt", "authentication"],
    "auto_generate_summary": true
}
```

Response: 201 Created

```
{
  "id": 157,
  "requester": { ... },
  "title": "How to implement JWT authentication in Django?",
  "kp_bounty": 200,
  "status": "open",
  "created_at": "2024-05-12T10:00:00Z",
  "ai_summary": "User is asking for help implementing JWT authentication..."
}
```

Errors: - 400 Bad Request: Invalid input (KP out of range, etc.) - 401 Unauthorized: Not authenticated - 402 Payment Required: Insufficient KP balance

3.3 Get Request Detail

Endpoint: GET /requests/{id}/

Response:

```
{
  "id": 1,
  "requester": { ... },
  "title": "...",
  "kp_bounty": 150,
  "status": "claimed",
  "claimed_by": {
    "id": 8,
    "username": "helper_dev",
    "trust_score": 82.0
  },
  "claimed_at": "2024-05-11T09:15:00Z",
  "solutions": [
    {
      "id": 12,
      "helper_id": 8,
      "solution_text": "Use select_related() for foreign keys...",
      "created_at": "2024-05-11T10:00:00Z",
      "is_approved": false
    }
  ],
  "chat_thread_id": 5,
  "activity_log": [...]
}
```

3.4 Claim Request

Endpoint: POST /requests/{id}/claim/

Response: 200 OK

```
{
  "status": "success",
  "message": "Request claimed successfully",
  "request_id": 1,
  "chat_thread_id": 5
}
```

Errors: - 400 Bad Request: Request already claimed - 404 Not Found: Request not found - 403 Forbidden: User suspended or low trust score

3.5 Submit Solution

Endpoint: POST /requests/{id}/submit-solution/

Request Body:

```
{
  "solution_text": "Here's how to optimize your queries...",
  "code_snippet": "# Python code\nMyModel.objects.select_related('foreign_key')..."
}
```

Response: 201 Created

```
{
  "solution_id": 12,
  "request_id": 1,
  "created_at": "2024-05-11T14:00:00Z",
  "status": "awaiting_review"
}
```

3.6 Rate Solution

Endpoint: POST /requests/{id}/rate-solution/

Request Body:

```
{
  "solution_id": 12,
  "rating": 5,
  "comment": "Exactly what I needed! Clear explanation.",
  "approve": true
}
```

Response: 200 OK

```
{
  "status": "success",
  "message": "Solution approved and KP transferred",
}
```

```
"kp_transferred": 150,  
"helper_id": 8,  
"request_status": "resolved"  
}
```

4. Freelance Jobs API

4.1 List Jobs

Endpoint: GET /jobs/

Query Parameters:

- page: integer
- status: string (open, assigned, in_progress, completed)
- min_budget: decimal
- max_budget: decimal
- sort_by: string (created, budget, deadline)

Response:

```
{  
  "count": 87,  
  "results": [  
    {  
      "id": 1,  
      "client": { ... },  
      "title": "Build API for IoT Dashboard",  
      "description": "Need Django REST API for...",  
      "budget_inr": 50000,  
      "status": "open",  
      "deadline": "2024-06-30T23:59:59Z",  
      "proposals_count": 5,  
      "milestones": [  
        {  
          "id": 1,  
          "title": "API Design & Setup",  
          "amount_inr": 15000,  
          "due_date": "2024-06-10"  
        }  
      ],  
      "created_at": "2024-05-12T00:00:00Z"  
    }  
  ]  
}
```

4.2 Create Job

Endpoint: POST /jobs/

Request Body:

```
{
  "title": "Build React Dashboard for Analytics",
  "description": "Need a React dashboard showing real-time analytics...",
  "budget_inr": 75000,
  "deadline": "2024-07-15",
  "milestones": [
    {
      "title": "UI Design & Setup",
      "amount_inr": 25000,
      "due_date": "2024-06-20"
    },
    {
      "title": "Backend Integration",
      "amount_inr": 30000,
      "due_date": "2024-07-10"
    },
    {
      "title": "Testing & Deployment",
      "amount_inr": 20000,
      "due_date": "2024-07-15"
    }
  ],
  "required_skills": ["react", "nodejs", "postgresql"]
}
```

Response: 201 Created

```
{
  "id": 88,
  "client_id": 3,
  "title": "Build React Dashboard...",
  "budget_inr": 75000,
  "status": "open",
  "milestones": [...]
}
```

4.3 Submit Proposal

Endpoint: POST /jobs/{id}/submit-proposal/

Request Body:

```
{
  "bid_amount_inr": 65000,
  "delivery_timeline_days": 30,
  "proposal_text": "I have 5+ years of React experience and completed similar projects..."
}
```

Response: 201 Created

```
{
  "proposal_id": 23,
  "job_id": 88,
  "contractor_id": 10,
  "status": "submitted"
}
```

4.4 Accept Proposal

Endpoint: POST /jobs/{id}/accept-proposal/

Request Body:

```
{
  "proposal_id": 23
}
```

Response: 200 OK

```
{
  "status": "success",
  "job_status": "assigned",
  "contractor_id": 10,
  "first_milestone_id": 1
}
```

4.5 Fund Milestone (Escrow)

Endpoint: POST /jobs/{id}/milestones/{milestone_id}/fund/

Creates Razorpay Order:

```
{
  "order_id": "order_...",
  "amount": 25000,
  "currency": "INR",
  "razorpay_key_id": "rzp_test_...",
  "client": {
    "name": "John Developer",
    "email": "john@example.com"
  }
}
```

Frontend handles Razorpay checkout → Server verifies payment

4.6 Submit Deliverable

Endpoint: POST /jobs/{id}/milestones/{milestone_id}/submit-deliverable/

Request Body:

```
{
  "submission_text": "Completed the API endpoints as per requirements...",
}
```

```
  "file_upload_url": "https://s3.../deliverable.zip"
}
```

Response: 201 Created

```
{
  "deliverable_id": 45,
  "milestone_id": 1,
  "status": "submitted",
  "awaiting_client_review": true
}
```

4.7 Approve Milestone (Release Escrow)

Endpoint: POST /jobs/{id}/milestones/{milestone_id}/approve/

Request Body:

```
{
  "approval_notes": "Excellent work! Exactly as specified.",
  "rating": 5
}
```

Response: 200 OK

```
{
  "status": "approved",
  "amount_released_inr": 25000,
  "contractor_wallet_updated": true,
  "next_milestone_id": 2
}
```

5. Payments API

5.1 Create Wallet Top-Up Order

Endpoint: POST /wallet/topup/

Request Body:

```
{
  "amount_inr": 5000
}
```

Response:

```
{
  "order_id": "order_...",
  "amount": 5000,
  "currency": "INR",
  "razorpay_key": "rzp_test_..."
}
```


5.2 Verify Payment & Credit Wallet

Endpoint: POST /wallet/topup/verify/

Request Body:

```
{
  "razorpay_order_id": "order_...",
  "razorpay_payment_id": "pay_...",
  "razorpay_signature": "..."}
}
```

Response: 200 OK

```
{
  "status": "success",
  "amount_credited": 5000,
  "new_balance": 12500,
  "transaction_id": "txn_..."
}
```

Errors: - 400 Bad Request: Invalid signature (fraud detected!) - 409 Conflict: Duplicate payment (idempotent check)

5.3 Get Wallet Balance

Endpoint: GET /wallet/balance/

Response:

```
{
  "wallet_inr": 12500.00,
  "knowledge_points": 350,
  "pending_withdrawals": 5000,
  "available_for_withdrawal": 7500
}
```

5.4 Request Withdrawal

Endpoint: POST /wallet/withdraw/

Request Body:

```
{
  "amount_inr": 5000,
  "bank_account_id": 1
}
```

Response: 201 Created

```
{
  "withdrawal_id": "wd_123",
  "amount_inr": 5000,
  "status": "pending",
}
```

```
    "expected_delivery": "2024-05-15"
}
```

6. Users API

6.1 Get User Profile

Endpoint: GET /users/{id}/

Response:

```
{
  "id": 5,
  "username": "john_dev",
  "email": "john@example.com",
  "bio": "Full-stack developer with 5+ years experience",
  "skills": ["django", "react", "postgresql"],
  "knowledge_points": 350,
  "trust_score": 78.5,
  "badges": ["helpful_contributor", "reliable_contractor"],
  "stats": {
    "requests_resolved": 42,
    "jobs_completed": 8,
    "average_rating": 4.7,
    "member_since": "2023-01-15"
  },
  "portfolio": {
    "featured_projects": [...],
    "achievements": [...]
  }
}
```

6.2 Update Profile

Endpoint: PATCH /users/me/

Request Body:

```
{
  "bio": "Updated bio",
  "skills": ["django", "react", "postgres", "kubernetes"],
  "notification_preference": "email"
}
```

Response: 200 OK

6.3 Get Current User

Endpoint: GET /users/me/

Returns: Full profile including private data (email, wallet, etc.)

6.4 Get User's Live Status

Endpoint: GET /users/me/live-status/

Response:

```
{
  "kp_balance": 350,
  "wallet_inr": 12500.00,
  "pending_requests": 3,
  "pending_jobs": 2,
  "unread_notifications": 5,
  "active_chat_threads": 7,
  "trust_score": 78.5,
  "is_suspended": false
}
```

7. Search API

7.1 Global Search

Endpoint: GET /search/

Query Parameters:

- q: string (required)
- category: string (requests, jobs, users)
- limit: integer (default: 50, max: 100)
- offset: integer (default: 0)

Request:

```
curl "https://helperlearner.com/api/v1/search/?q=django%20authentication&category=requests"
```

Response:

```
{
  "results": {
    "requests": [
      {
        "id": 1,
        "type": "help_request",
        "title": "How to implement Django authentication?",
        "score": 0.95
      }
    ],
    "jobs": [
      {
        "id": 88,
        "type": "freelance_job",
        "title": "Build API with Django authentication",

```

```

        "score": 0.87
    }
],
"users": [
    {
        "id": 5,
        "type": "user",
        "username": "django_expert",
        "score": 0.72
    }
]
}
}

```

8. Notification API

8.1 List Notifications

Endpoint: GET /notifications/

Query Parameters:

- unread_only: boolean (default: false)
- limit: integer (default: 20)

Response:

```

{
  "unread_count": 3,
  "results": [
    {
      "id": 101,
      "type": "request_claimed",
      "title": "Your request was claimed",
      "message": "john_dev has claimed your Django ORM question",
      "related_object": "/api/v1/requests/1/",
      "created_at": "2024-05-12T10:00:00Z",
      "is_read": false
    }
  ]
}

```

8.2 Mark as Read

Endpoint: POST /notifications/{id}/read/

Response: 204 No Content

9. Chat API

9.1 Get Chat Thread

Endpoint: GET /chat/threads/{id}/

Response:

```
{
  "id": 5,
  "request_id": 1,
  "subject": "Django ORM Query Optimization",
  "participants": [...],
  "messages": [
    {
      "id": 23,
      "sender": { ... },
      "content": "How to optimize N+1 queries?",
      "created_at": "2024-05-11T10:00:00Z",
      "is_read": true
    }
  ]
}
```

9.2 Send Message

Endpoint: POST /chat/threads/{id}/messages/

Request Body:

```
{
  "content": "Try using select_related() for this query",
  "mentions": [5, 8]
}
```

Response: 201 Created (WebSocket broadcasts to other participants in real-time)

10. Error Handling

Standard Error Response

```
{
  "error": true,
  "error_code": "INSUFFICIENT_KP",
  "message": "You don't have enough KP to post this request",
  "details": {
    "required_kp": 150,
    "available_kp": 100
  },
  "timestamp": "2024-05-12T10:30:00Z"
}
```

Common Error Codes

Code	HTTP Status	Message
INVALID_INPUT	400	Request body validation failed
UNAUTHORIZED	401	Authentication required
INSUFFICIENT_KP	402	Not enough Knowledge Points
FORBIDDEN	403	You don't have permission
NOT_FOUND	404	Resource not found
CONFLICT	409	Request conflicts with current state
RATE_LIMITED	429	Too many requests
SERVER_ERROR	500	Internal server error

11. Pagination

All list endpoints support pagination:

?page=1&page_size=20

Response includes:

```
{
  "count": 156,
  "next": "https://...?page=2",
  "previous": null,
  "results": [...]
}
```

12. WebSocket Events (Real-time)

Connection: wss://helperlearner.com/ws/chat/{thread_id}/

Events:

```
{
  "type": "message",
  "data": {
    "id": 23,
    "sender_id": 5,
    "content": "...",
    "created_at": "2024-05-12T10:00:00Z"
  }
}
```

Broadcast Events: - message_sent: New chat message - request_claimed: Help request claimed
- milestone_approved: Job milestone approved (funds released) - notification_created: In-app notification

Complete API Reference Summary

Method	Endpoint	Auth	Description
GET	/requests/	Any	List help requests
POST	/requests/	Auth	Create help request
GET	/requests/{id}/	Any	Get request detail
POST	/requests/{id}/claim/	Auth	Claim request
POST	/requests/{id}/submit-solution/	Auth	Submit solution
POST	/requests/{id}/rate-solution/	Auth	Rate solution
GET	/jobs/	Any	List freelance jobs
POST	/jobs/	Auth	Create job
POST	/jobs/{id}/submit-proposal/	Auth	Submit proposal
POST	/jobs/{id}/accept-proposal/	Auth	Accept proposal
POST	/jobs/{id}/milestones/{mid}/fund/	Auth	Fund milestone (Razorpay)
POST	/wallet/topup/	Auth	Create top-up order
POST	/wallet/topup/verify/	Auth	Verify & credit wallet
GET	/users/{id}/	Any	Get user profile
GET	/users/me/	Auth	Get current user
GET	/search/	Any	Global search
GET	/notifications/	Auth	List notifications
POST	/chat/threads/{id}/message/	Auth	Send message

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